

Q1. A helium-neon laser usually used in supermarket scanners emits light of wavelength 633 nm. Calculate the frequency of this light. What is the energy of a photon generated by this laser?

A1

$$\lambda = 633 \text{ nm}$$

$$\gamma = \frac{c}{\lambda} = \frac{3 \times 10^8}{633 \times 10^{-9}} \text{ s}^{-1}$$

$$= 4.74 \times 10^{14} \text{ s}^{-1}$$

$$= 4.74 \times 10^{14} \text{ Hz}$$

$$E = h\gamma = 6.624 \times 10^{-34} \text{ J.s} \times 4.74 \times 10^{14} \text{ Hz}$$

$$= 3.14 \times 10^{-19} \text{ J}$$

$$E = \cancel{h}\gamma = \frac{\cancel{h}c}{\lambda} = hc\tilde{\nu}$$

Q2. When lithium is irradiated with light, the kinetic energy of the ejected electrons is given as:
 $2.935 \times 10^{-19} \text{ J}$ for $\lambda = 300.0 \text{ nm}$, and $1.280 \times 10^{-19} \text{ J}$ for $\lambda = 400.0 \text{ nm}$. Calculate the (a)
 Planck constant, (b) work function, and (b) threshold frequency of lithium from these data.

Ans (a) $KE_1 = h\nu_1 - \phi \quad // \lambda_1 = 300 \text{ nm}$
 $KE_2 = h\nu_2 - \phi \quad // \lambda_2 = 400 \text{ nm}$

$$KE_1 - KE_2 = h(\nu_1 - \nu_2) = hc \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right)$$

$$\Rightarrow (2.935 - 1.280) \times 10^{-19} \text{ J} = hc \left(\frac{1}{300 \text{ nm}} - \frac{1}{400 \text{ nm}} \right)$$

$$\Rightarrow 1.655 \times 10^{-19} \text{ J} =$$

$$h (2.998 \times 10^8 \text{ m s}^{-1}) \times \left(\frac{1}{300 \times 10^{-9} \text{ m}} - \frac{1}{400 \times 10^{-9} \text{ m}} \right)$$

$$\Rightarrow h = \frac{1.655 \times 10^{-19} \text{ J}}{2.998 \times 10^8 \text{ m s}^{-1}} \times \frac{300 \times 400 \times 10^{-9} \text{ m}}{(400 - 300)}$$

$$= 6.624 \times 10^{-34} \text{ J.s}$$

(b) Using $\lambda = 300 \text{ nm}$

$$2.935 \times 10^{-19} \text{ J} = \frac{h \times c}{300 \times 10^{-9} \text{ m}} - h\nu_0$$

$$\Rightarrow h\nu_0 = \frac{h c}{300 \times 10^{-9} \text{ m}} - 2.935 \times 10^{-19} \text{ J}$$
$$= \frac{6.626 \times 10^{-34} \text{ J.s} \times 2.998 \times 10^8 \text{ m/s}}{300 \times 10^{-9} \text{ m}} - 2.935 \times 10^{-19} \text{ J}$$

$$\Rightarrow h\nu_0 = 3.6865 \times 10^{-19} \text{ J} \quad \text{Work function}$$
$$= 2.301 \text{ eV} \quad [\text{NOTE : } 1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}]$$

(c)

$$\Rightarrow \nu_0 = \frac{3.6865 \times 10^{-19} \text{ J}}{6.626 \times 10^{-34} \text{ J.s}}$$

$$\Rightarrow \nu_0 = 5.564 \times 10^{14} \text{ Hz} \quad \text{threshold frequency}$$

Q3. Calculate the de Broglie wavelength for a baseball (0.14 kg) traveling at 144 km/h. Now find out the de Broglie wavelength of an electron traveling at 1.0 % of the speed of light. Interpret your results on these two moving objects.

[Note: Mass of an electron = 9.109×10^{-31} kg]

A3.

Baseball

Weight

0.14 kg

Speed

144 km/h

mv
momentum
(p)

$$0.14 \text{ kg} \times 144 \text{ km/h}$$

$$= 0.14 \text{ kg} \times \frac{144 \times 10^3}{3600}$$

$$= 5.6 \text{ kg m s}^{-1}$$

de Broglie wavelength

$$\lambda = \frac{h}{p}$$

$$= \frac{6.626 \times 10^{-34} \text{ J s}}{5.6 \text{ kg m s}^{-1}}$$

$$= \frac{6.626 \times 10^{-34} \text{ kg m s}^{-2} \text{ s}}{5.6 \text{ kg m s}^{-1}}$$

$$= 1.2 \times 10^{-34} \text{ m}$$

so small to be detectable
practically

Electron

$$9.109 \times 10^{-31} \text{ kg}$$

$$c \times 1.0 \%$$

$$= 2.998 \times 10^8 \text{ m s}^{-1} \times \frac{1}{100}$$

$$= 2.998 \times 10^6 \text{ m s}^{-1}$$

$$9.109 \times 10^{-31} \text{ kg} \times 2.998 \times 10^6 \text{ m s}^{-1}$$

$$= 9.109 \times 10^{-31} \times 2.998 \times 10^6$$

$$= 2.73 \times 10^{-24} \text{ kg m s}^{-1}$$

$$\lambda = \frac{h}{p}$$

$$= \frac{6.626 \times 10^{-34} \text{ J s}}{2.73 \times 10^{-24} \text{ kg m s}^{-1}}$$

$$= \frac{6.626 \times 10^{-34} \text{ kg m s}^{-2} \text{ s}}{2.73 \times 10^{-24} \text{ kg m s}^{-1}}$$

$$= 2.43 \times 10^{-10} \text{ m}$$

reasonable as of atomic
dimension